



BobSim

SteadyStateEval Vehicle Characterization

Measured- a_y ramp-steer isolines with robust fits

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Run Notes

- Open-loop ramp-steer at 12.5, 15, 17.5, and 20 m/s
- steerStart = 2.0 s; trim all pre-ramp samples
- Measured a_y is the analysis axis; commanded a_y only sets the ramp
- lmeasured a_y < 1.0 m/s² is excluded; linear band = 1.0 to 4.0 m/s²
- Understeer gradient = mean $d(\text{delta_excess})/d(a_y)$ over the linear band
- Response curves use split Hampel-filtered spline fits; sensitivities trim the outer 20% of lmeasured a_y
- steerExcess = avgSteerAngle - wheelbase * curvature
- Tire/config: MF52, baseline_v3; Backend: OpenModelica

SteadyStateEval Summary 12.5 m/s

Metric	Value	Units
Measured a_y Range	$-16.97 \rightarrow 17.37$	m/s^2
Roadwheel Angle Gradient	0.01029	$\frac{\text{rad}}{\text{m/s}^2}$
	5.78	$\frac{\text{deg}}{g}$
Handwheel Angle Gradient	0.04223	$\frac{\text{rad}}{\text{m/s}^2}$
	23.74	$\frac{\text{deg}}{g}$
Sideslip Gradient	0.00199	$\frac{\text{rad}}{\text{m/s}^2}$
	1.12	$\frac{\text{deg}}{g}$
Understeer Gradient	0.00072	$\frac{\text{rad}}{\text{m/s}^2}$
	0.40	$\frac{\text{deg}}{g}$
Handwheel Understeer Gradient	0.04223	$\frac{\text{rad}}{\text{m/s}^2}$
	23.74	$\frac{\text{deg}}{g}$
Roll Gradient	1.414	$\frac{\text{deg}}{g}$
Handwheel Torque Min	-18.2	$\text{N} \cdot \text{m}$
Handwheel Torque Max	17.5	$\text{N} \cdot \text{m}$

SteadyStateEval Summary 15.0 m/s

Metric	Value	Units
Measured a_y Range	$-17.33 \rightarrow 18.05$	m/s^2
Roadwheel Angle Gradient	0.00761	$\frac{\text{rad}}{\text{m/s}^2}$
	4.28	$\frac{\text{deg}}{g}$
Handwheel Angle Gradient	0.03127	$\frac{\text{rad}}{\text{m/s}^2}$
	17.58	$\frac{\text{deg}}{g}$
Sideslip Gradient	0.00078	$\frac{\text{rad}}{\text{m/s}^2}$
	0.44	$\frac{\text{deg}}{g}$
Understeer Gradient	0.00076	$\frac{\text{rad}}{\text{m/s}^2}$
	0.43	$\frac{\text{deg}}{g}$
Handwheel Understeer Gradient	0.03127	$\frac{\text{rad}}{\text{m/s}^2}$
	17.58	$\frac{\text{deg}}{g}$
Roll Gradient	1.411	$\frac{\text{deg}}{g}$
Handwheel Torque Min	-17.5	$\text{N} \cdot \text{m}$
Handwheel Torque Max	16.7	$\text{N} \cdot \text{m}$

SteadyStateEval Summary 17.5 m/s

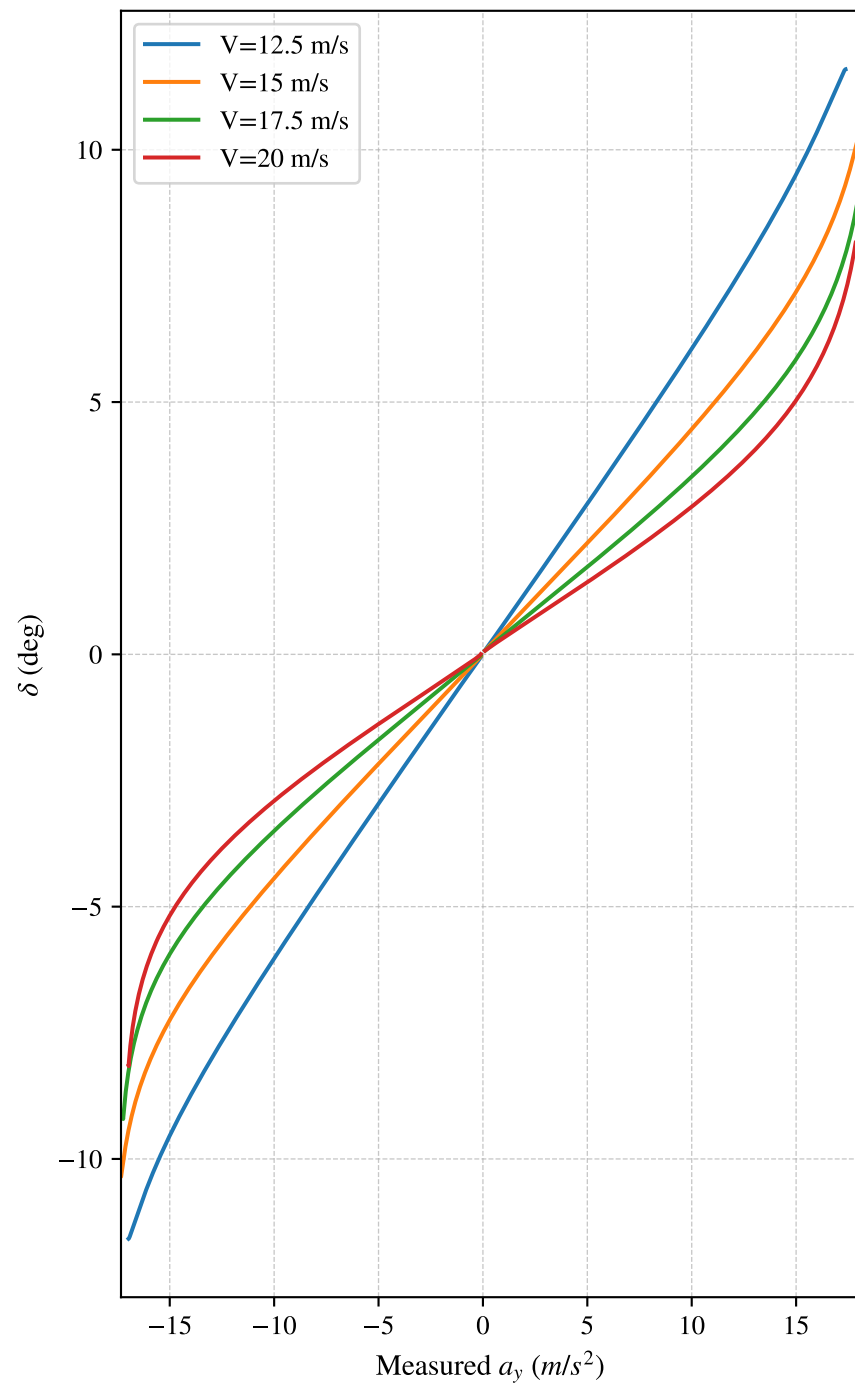
Metric	Value	Units
Measured a_y Range	$-17.23 \rightarrow 18.05$	m/s^2
Roadwheel Angle Gradient	0.00593	$\frac{\text{rad}}{\text{m/s}^2}$
	3.34	$\frac{\text{deg}}{g}$
Handwheel Angle Gradient	0.02437	$\frac{\text{rad}}{\text{m/s}^2}$
	13.70	$\frac{\text{deg}}{g}$
Sideslip Gradient	0.00006	$\frac{\text{rad}}{\text{m/s}^2}$
	0.03	$\frac{\text{deg}}{g}$
Understeer Gradient	0.00078	$\frac{\text{rad}}{\text{m/s}^2}$
	0.44	$\frac{\text{deg}}{g}$
Handwheel Understeer Gradient	0.02437	$\frac{\text{rad}}{\text{m/s}^2}$
	13.70	$\frac{\text{deg}}{g}$
Roll Gradient	1.421	$\frac{\text{deg}}{g}$
Handwheel Torque Min	-17.0	$\text{N} \cdot \text{m}$
Handwheel Torque Max	16.2	$\text{N} \cdot \text{m}$

SteadyStateEval Summary 20.0 m/s

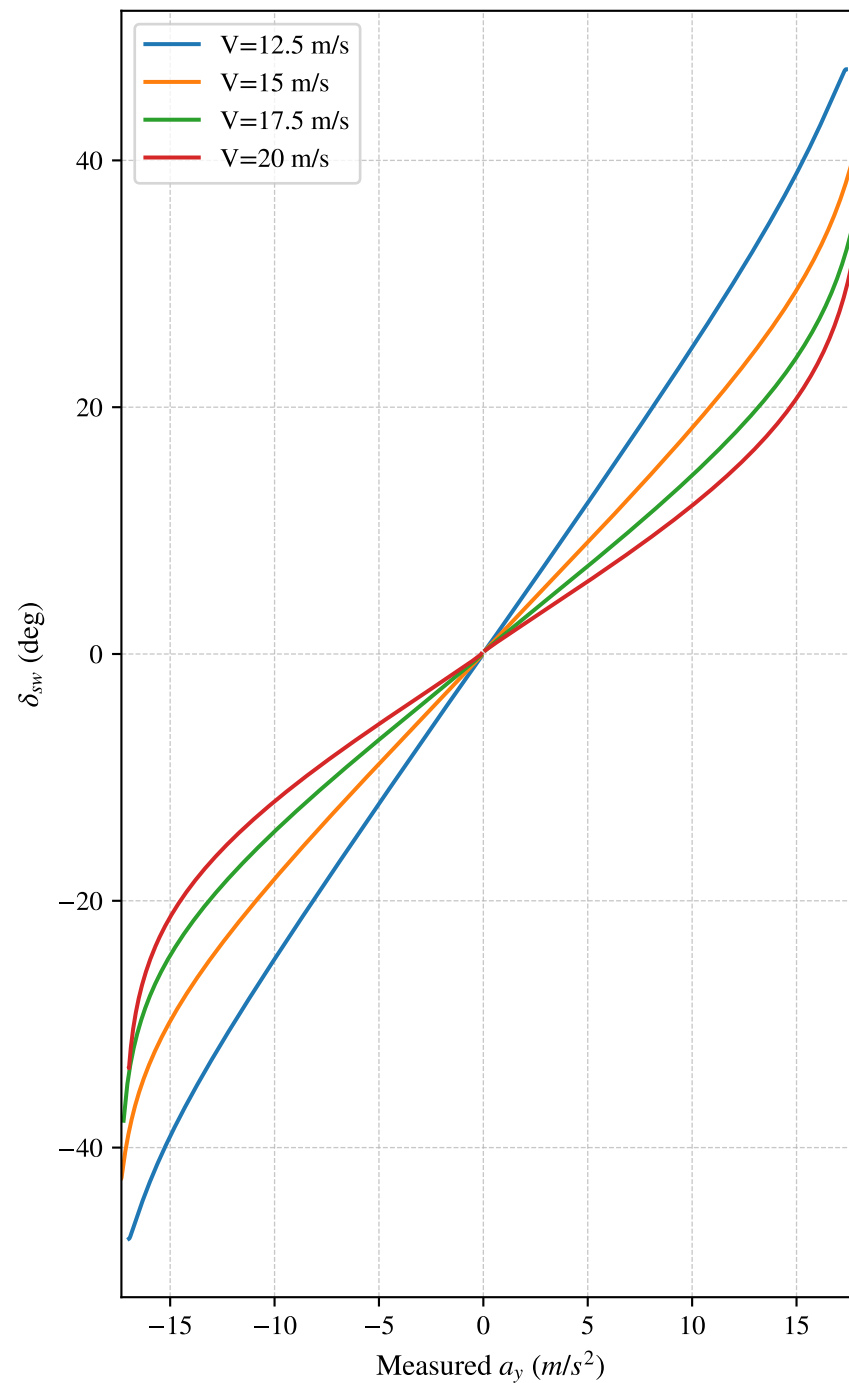
Metric	Value	Units
Measured a_y Range	$-16.97 \rightarrow 17.86$	m/s^2
Roadwheel Angle Gradient	0.00500	$\frac{\text{rad}}{\text{m/s}^2}$
	2.81	$\frac{\text{deg}}{g}$
Handwheel Angle Gradient	0.02053	$\frac{\text{rad}}{\text{m/s}^2}$
	11.54	$\frac{\text{deg}}{g}$
Sideslip Gradient	-0.00040	$\frac{\text{rad}}{\text{m/s}^2}$
	-0.23	$\frac{\text{deg}}{g}$
Understeer Gradient	0.00097	$\frac{\text{rad}}{\text{m/s}^2}$
	0.54	$\frac{\text{deg}}{g}$
Handwheel Understeer Gradient	0.02053	$\frac{\text{rad}}{\text{m/s}^2}$
	11.54	$\frac{\text{deg}}{g}$
Roll Gradient	1.401	$\frac{\text{deg}}{g}$
Handwheel Torque Min	-16.6	$\text{N} \cdot \text{m}$
Handwheel Torque Max	15.8	$\text{N} \cdot \text{m}$

Steering Angle Isolines vs Measured a_y

Roadwheel Angle δ

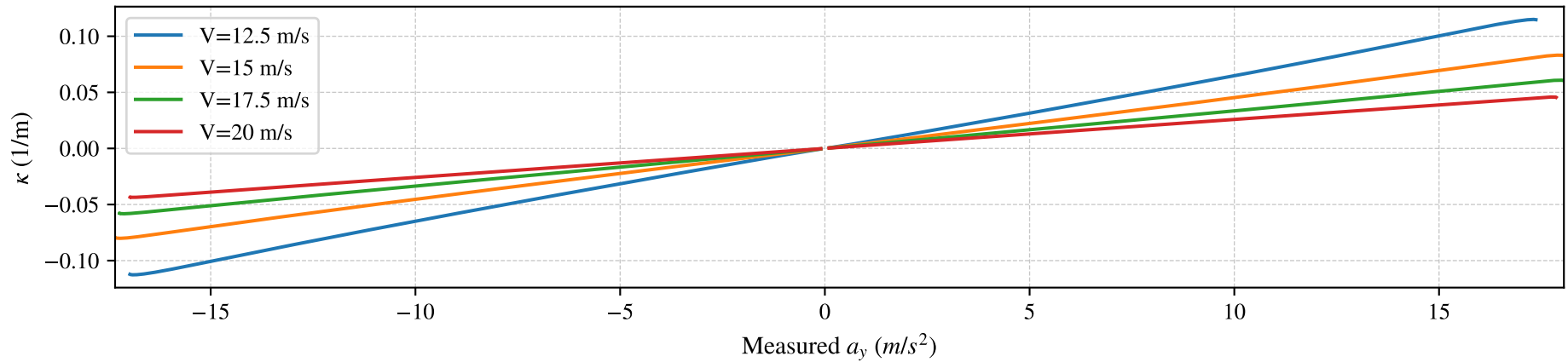


Handwheel Angle δ_{sw}

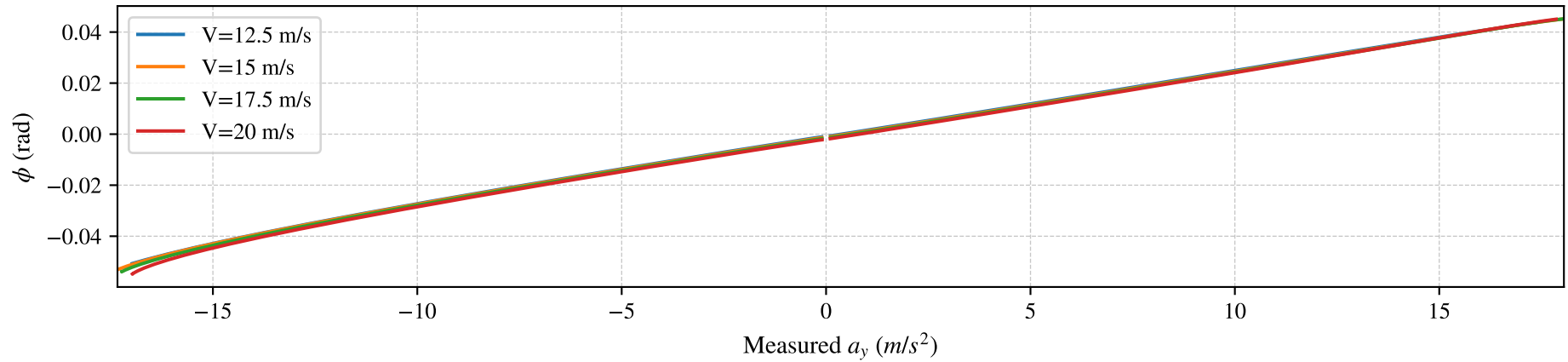


Vehicle State Isolines vs Measured a_y

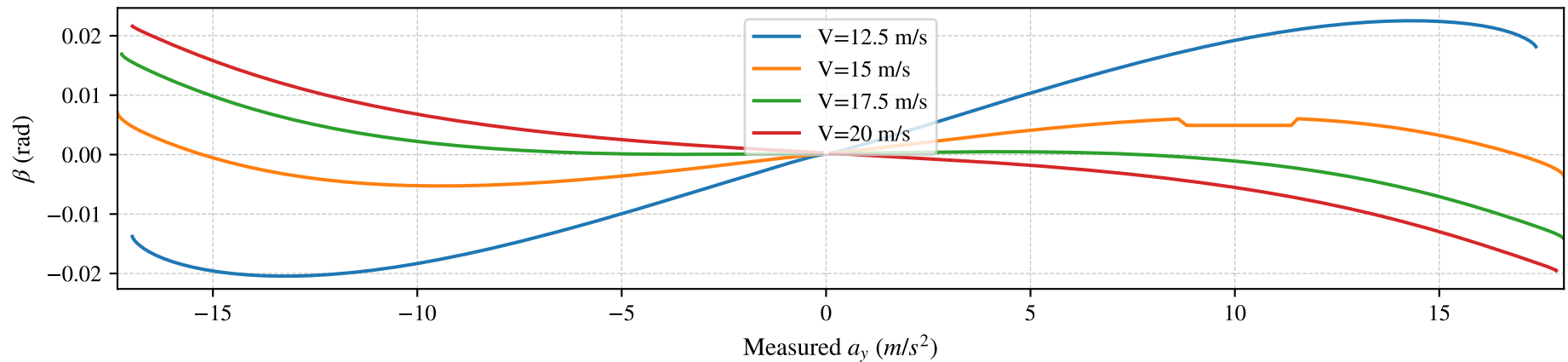
Curvature κ



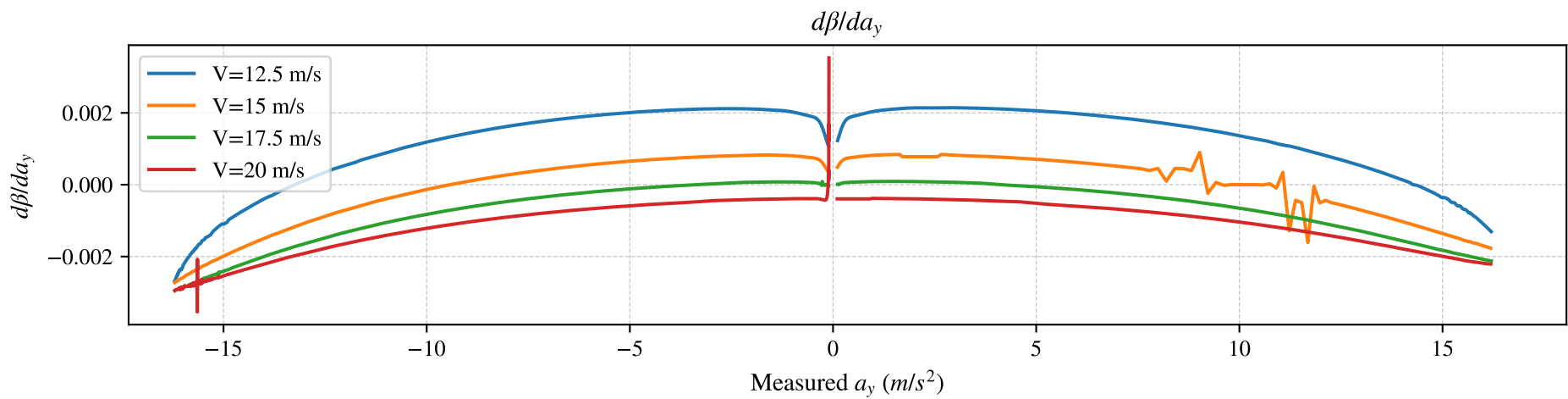
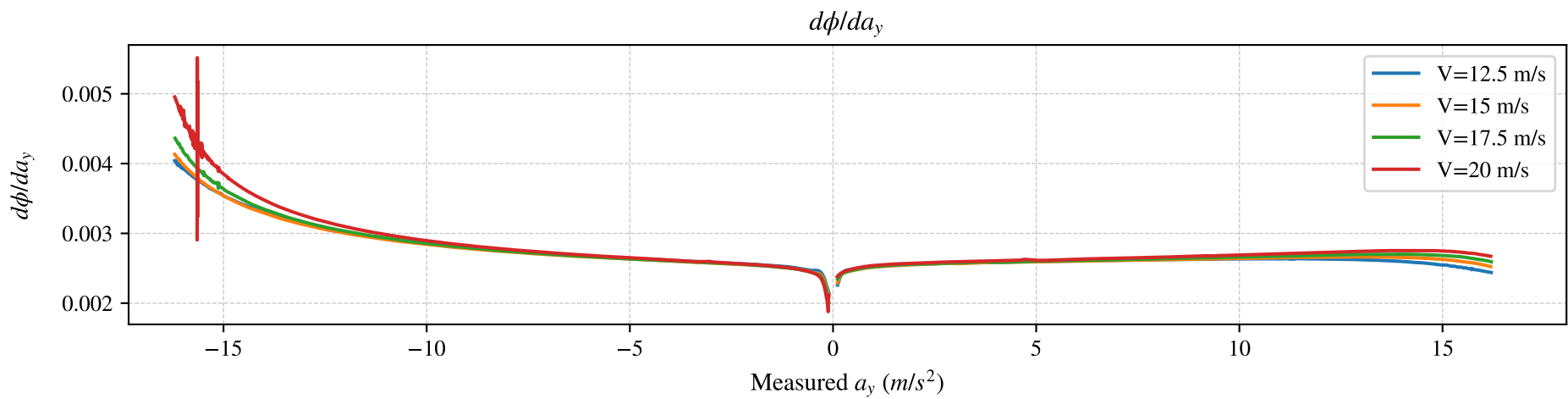
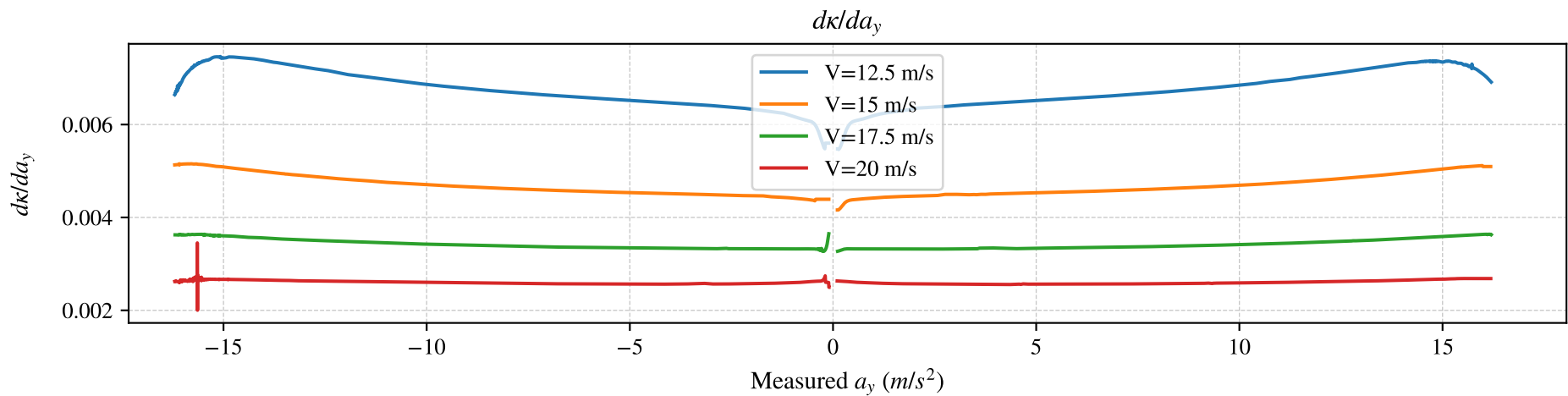
Roll Angle ϕ



Sideslip β

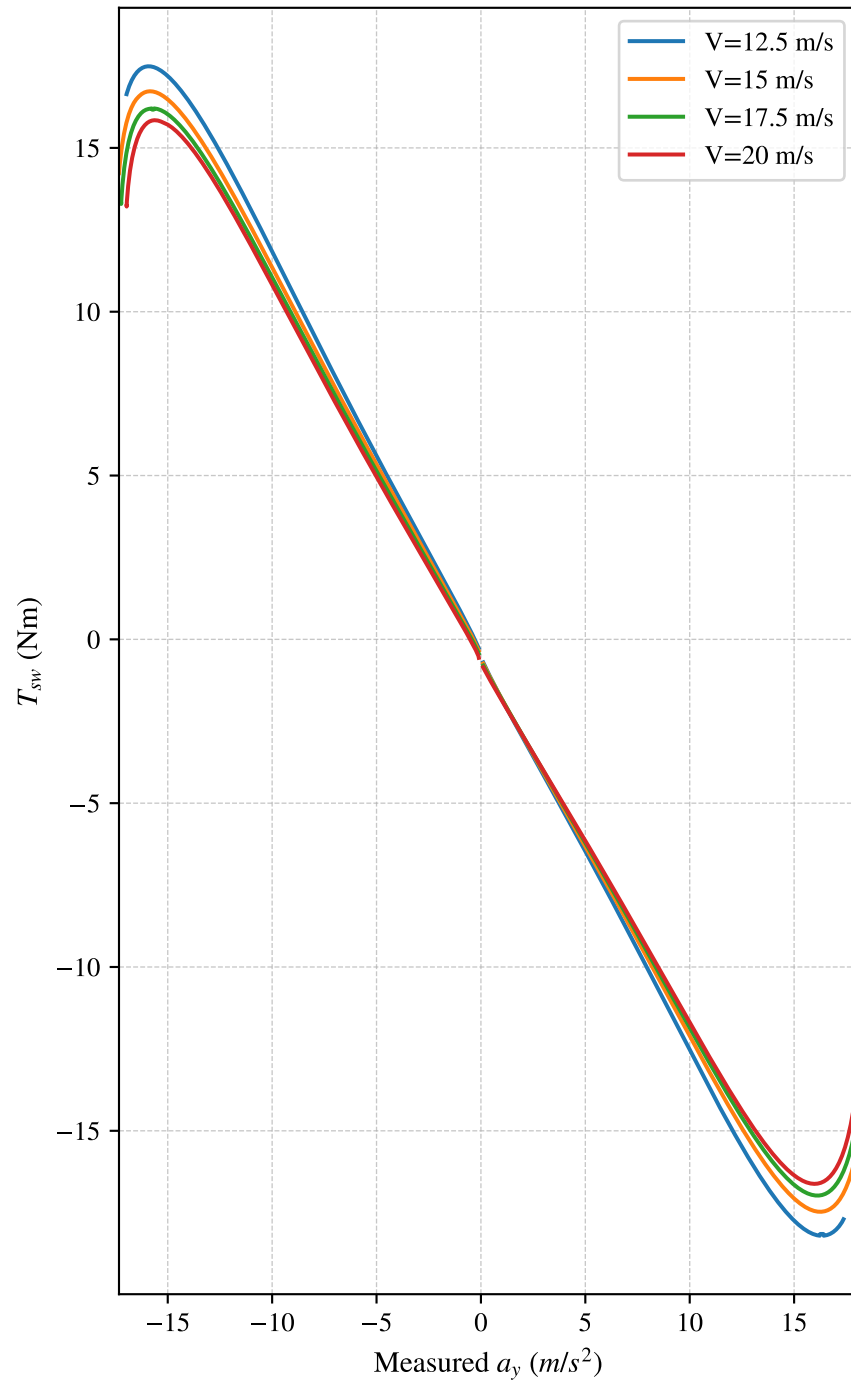


Parametric Response Sensitivities vs Measured a_y

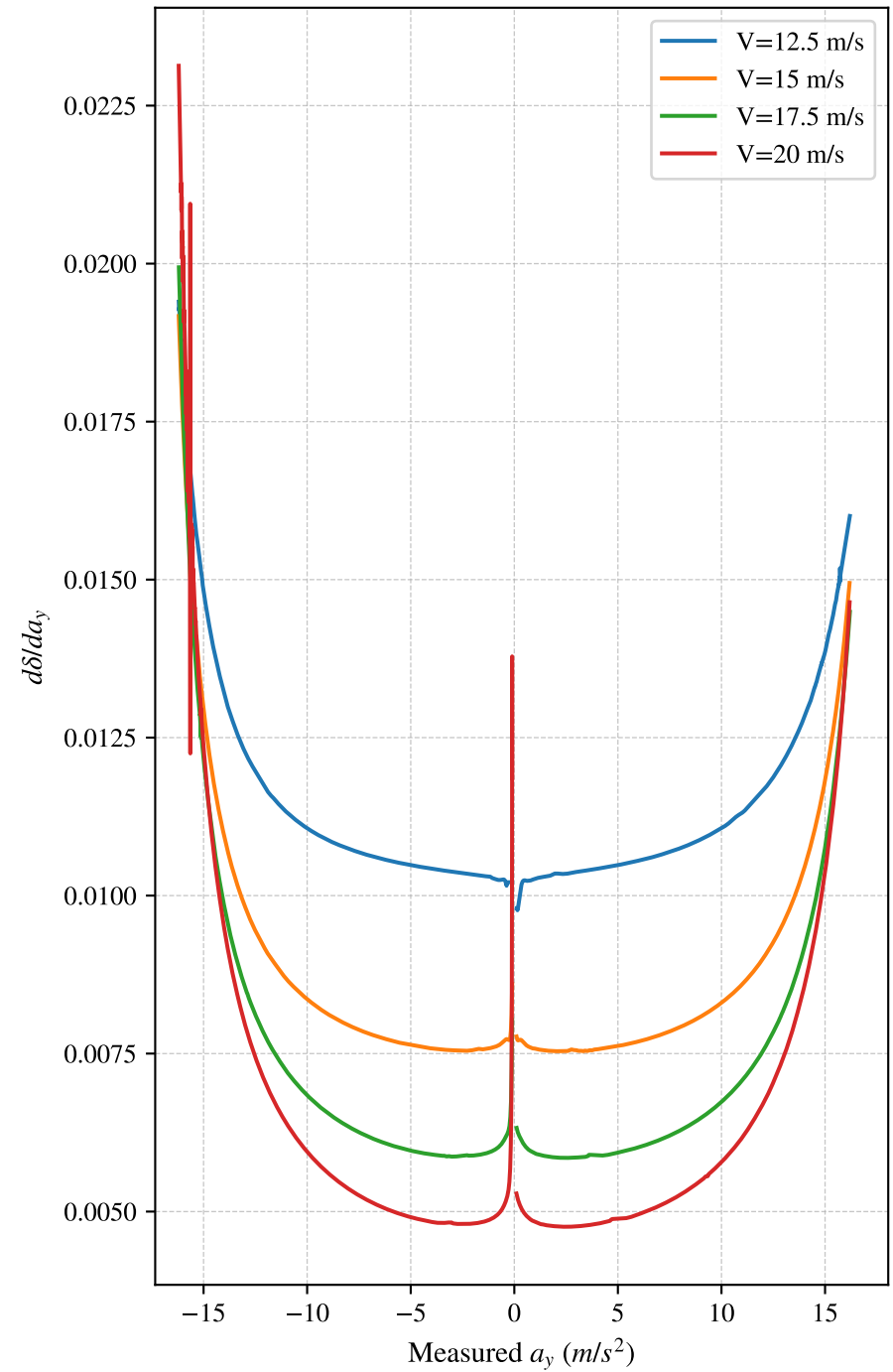


Steering Effort Isolines vs Measured a_y

Steering Torque T_{sw}

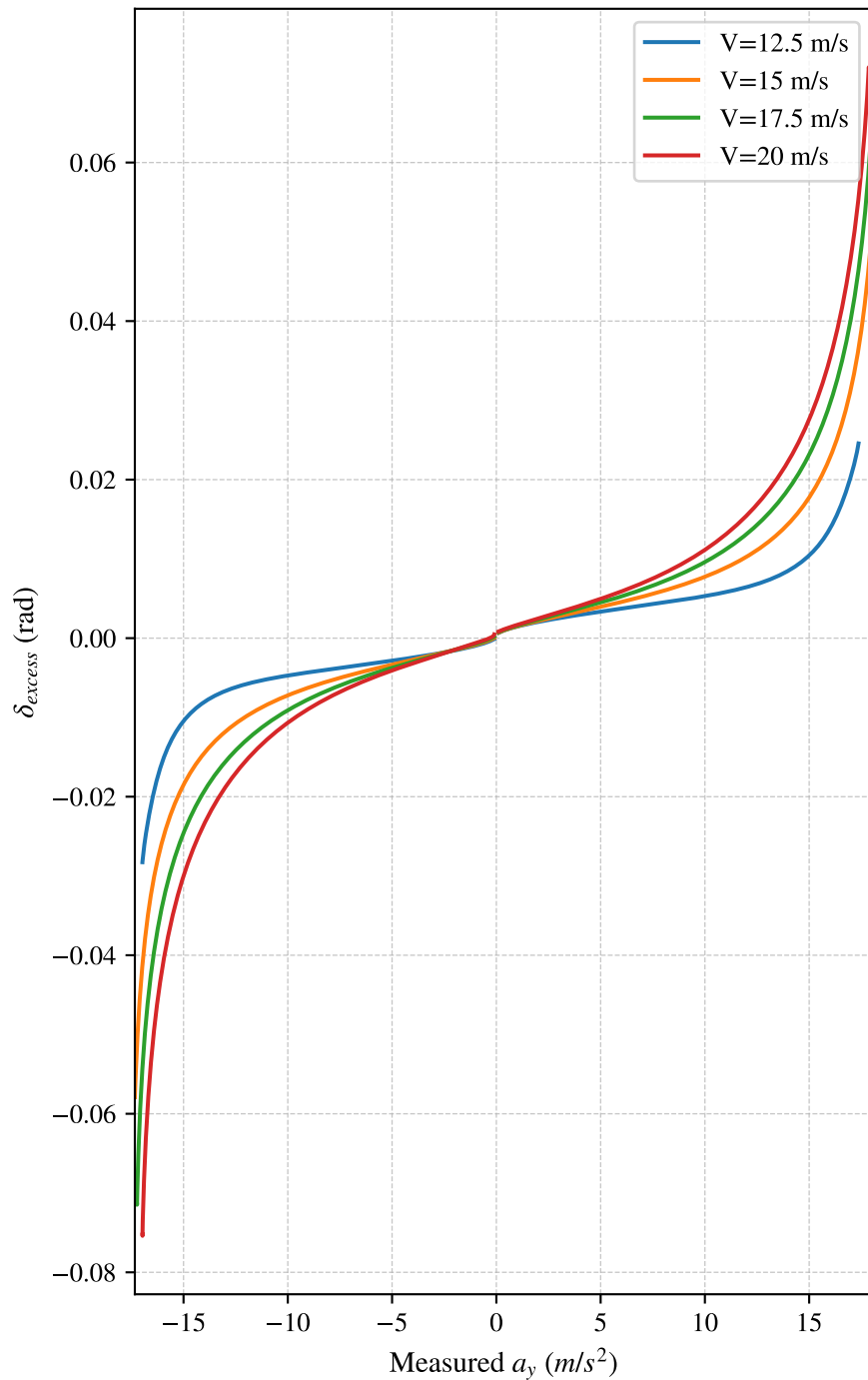


Roadwheel Steering Sensitivity $d\delta/da_y$

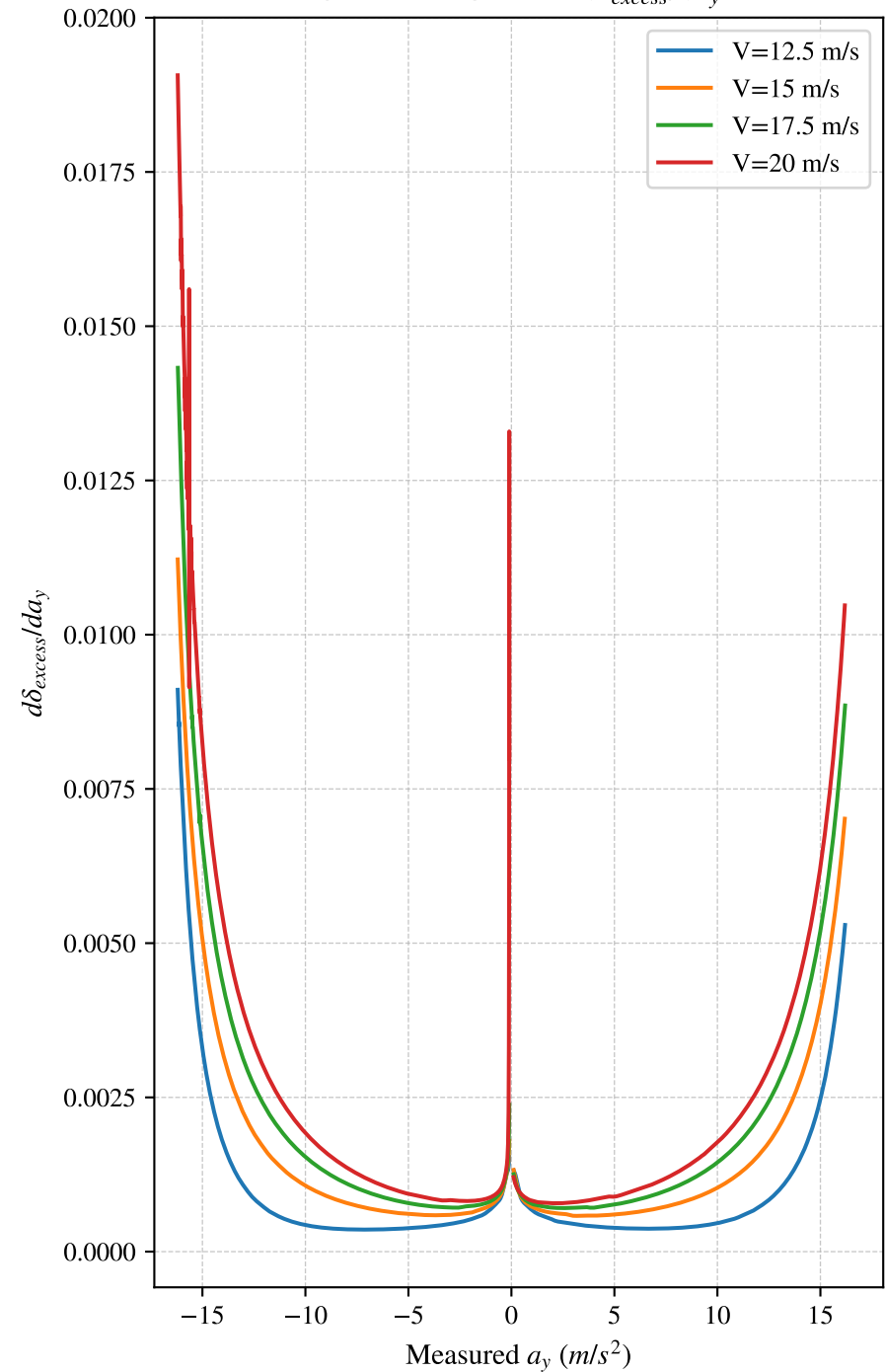


Excess Steer Isolines vs Measured a_y

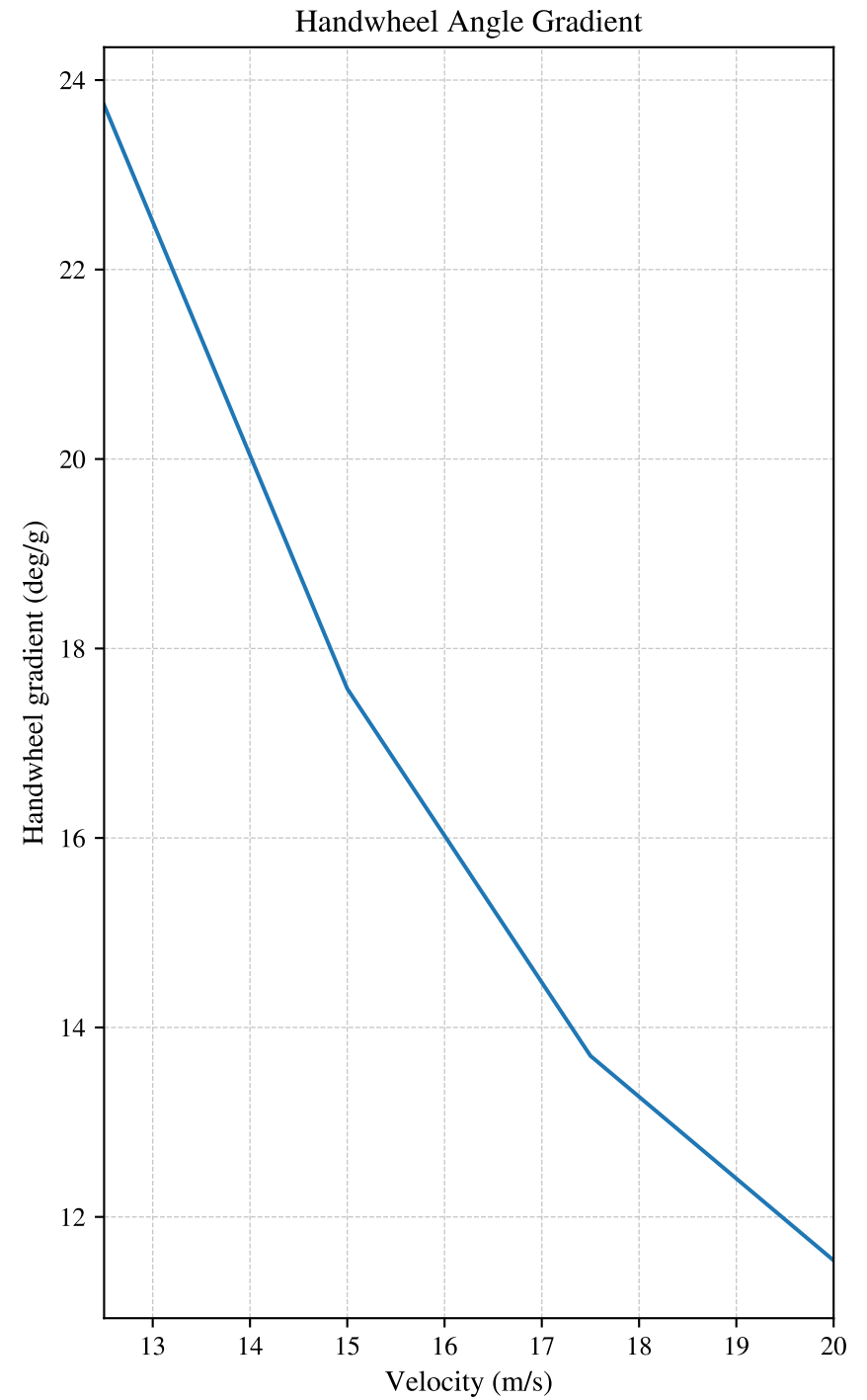
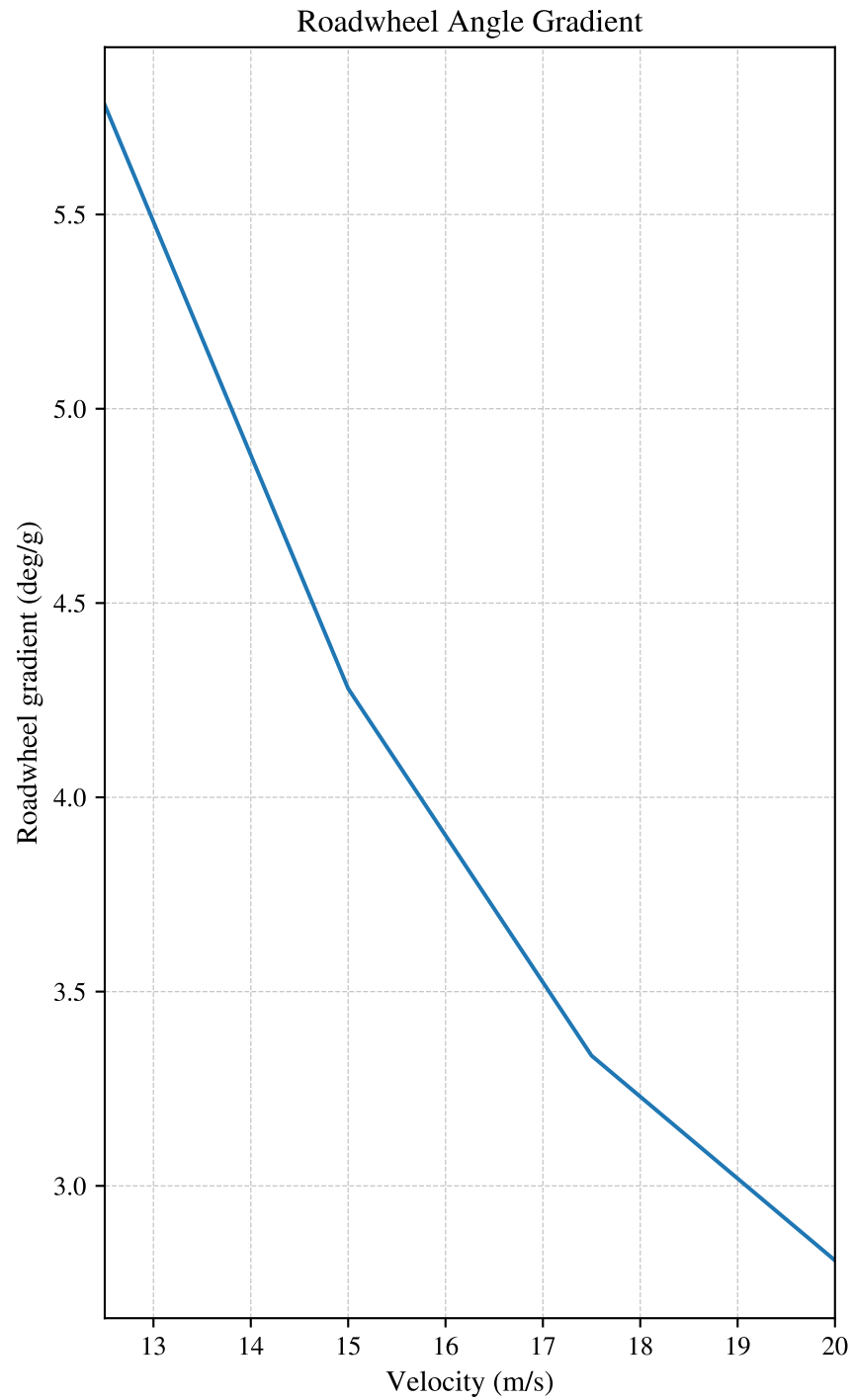
Steer Excess δ_{excess}



Understeer Gradient $d\delta_{excess}/da_y$

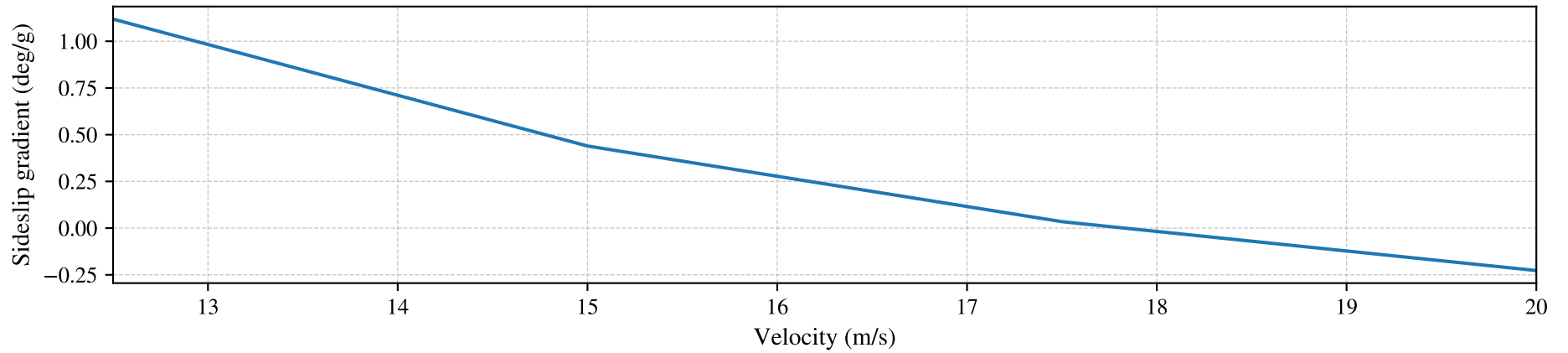


Angle Gradients vs Velocity

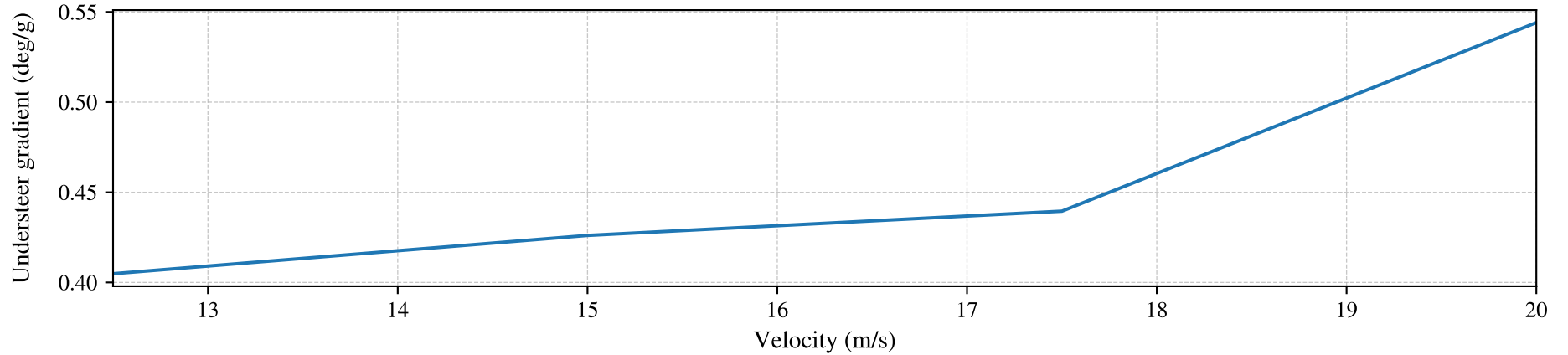


State Gradients vs Velocity

Sideslip Gradient



Understeer Gradient



Roll Gradient

